Exploratory Data Analysis on the Automobile Dataset

Report

Introduction

This report explores the cleaned automobile dataset by removing redundant columns, duplicate rows, and missing values before investigating the relationships between vehicle characteristics and price.

Data cleaning

• Removed redundant columns: normalized-losses and symboling.
• Removed duplicate rows.
• Removed rows with missing values.
• Converted numeric fields to correct data types.

Missing data

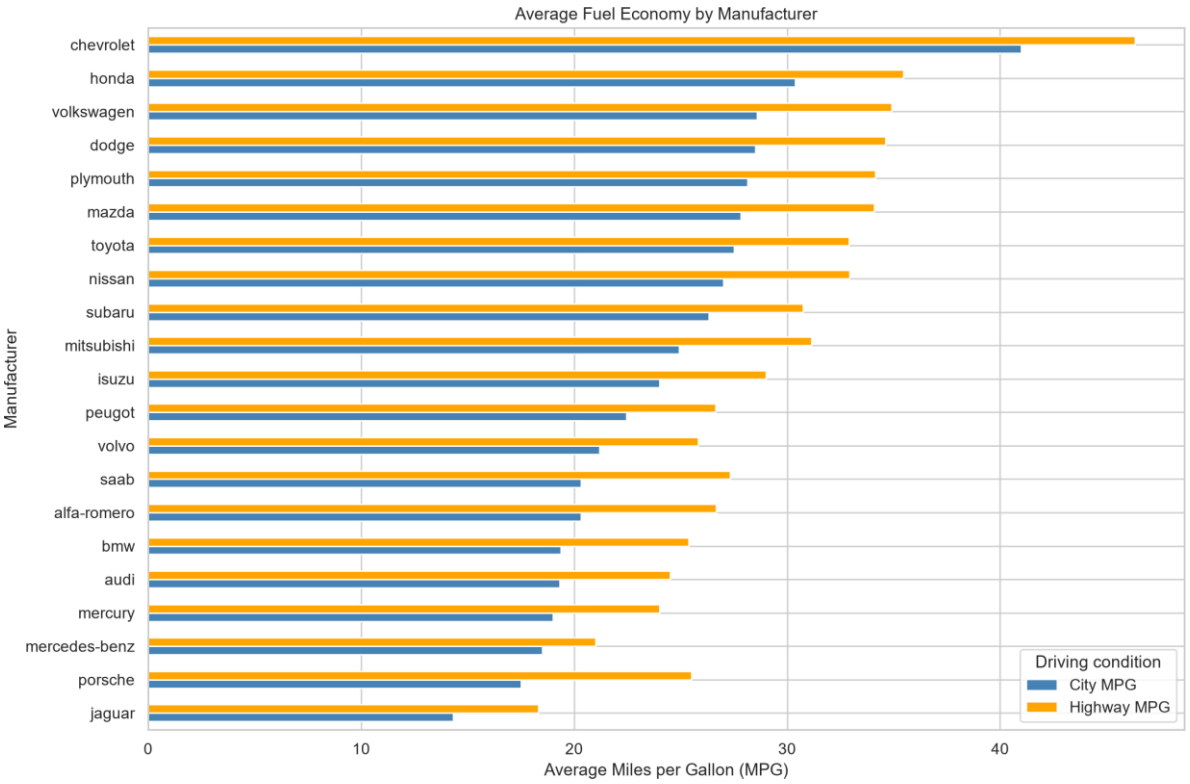
The original Automobile dataset contained missing values represented by the symbol ? instead of standard null values. These missing values appeared in several attributes, including normalized-losses, num-of-doors, bore, stroke, horsepower, peak-rpm, and price. Before analysis, the ? values were identified as missing data. The normalized-losses column was removed because it contained many missing values and was not required for this analysis. Records containing missing values in the remaining variables were removed to produce a complete dataset for exploratory data analysis. After data cleaning, the dataset contained 193 observations and 24 variables, ensuring that all subsequent visualizations and statistical analyses were based on complete and consistent data.

Data stories and visualisations

The following visualisations answer every exploration question in the notebook. Each chart is generated by executable code in the submitted notebook.

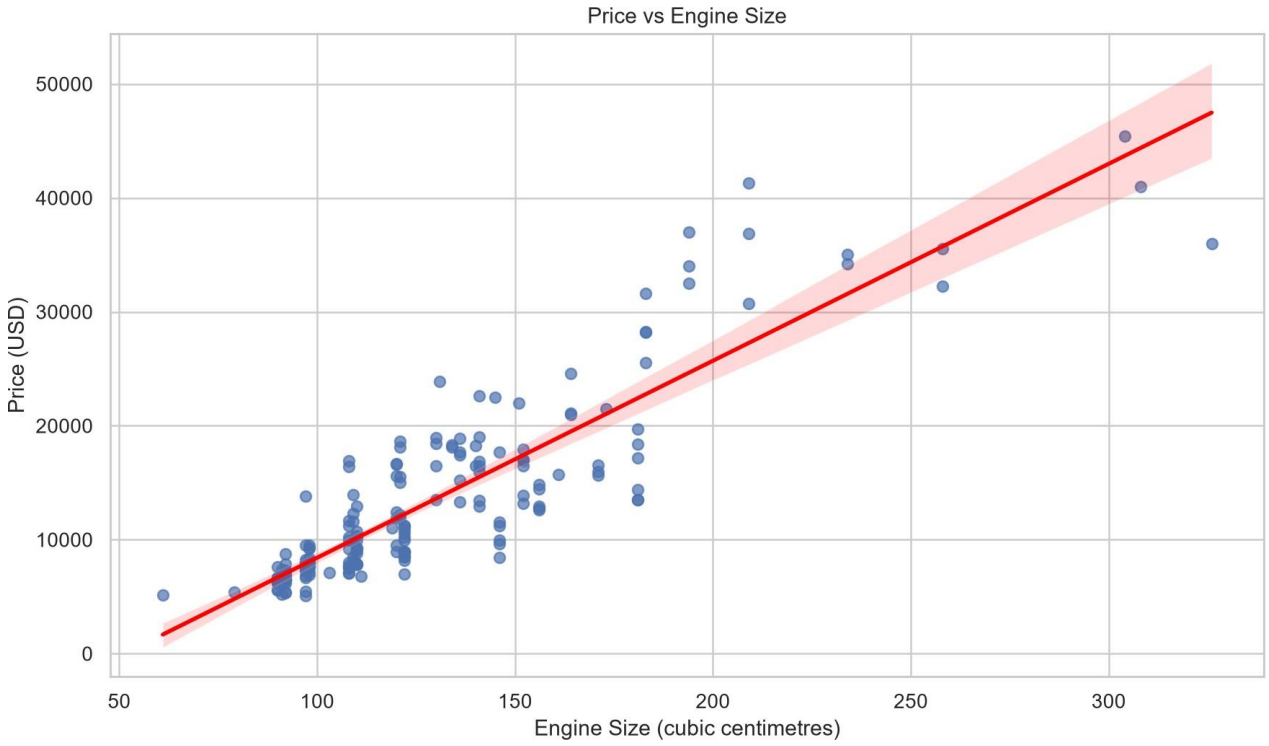
Visualisation 1: Average City MPG by Manufacturer

Chevrolet has the highest average city fuel economy. The chart also compares highway MPG, while manufacturers such as Jaguar and Mercedes-Benz generally have lower city MPG values.



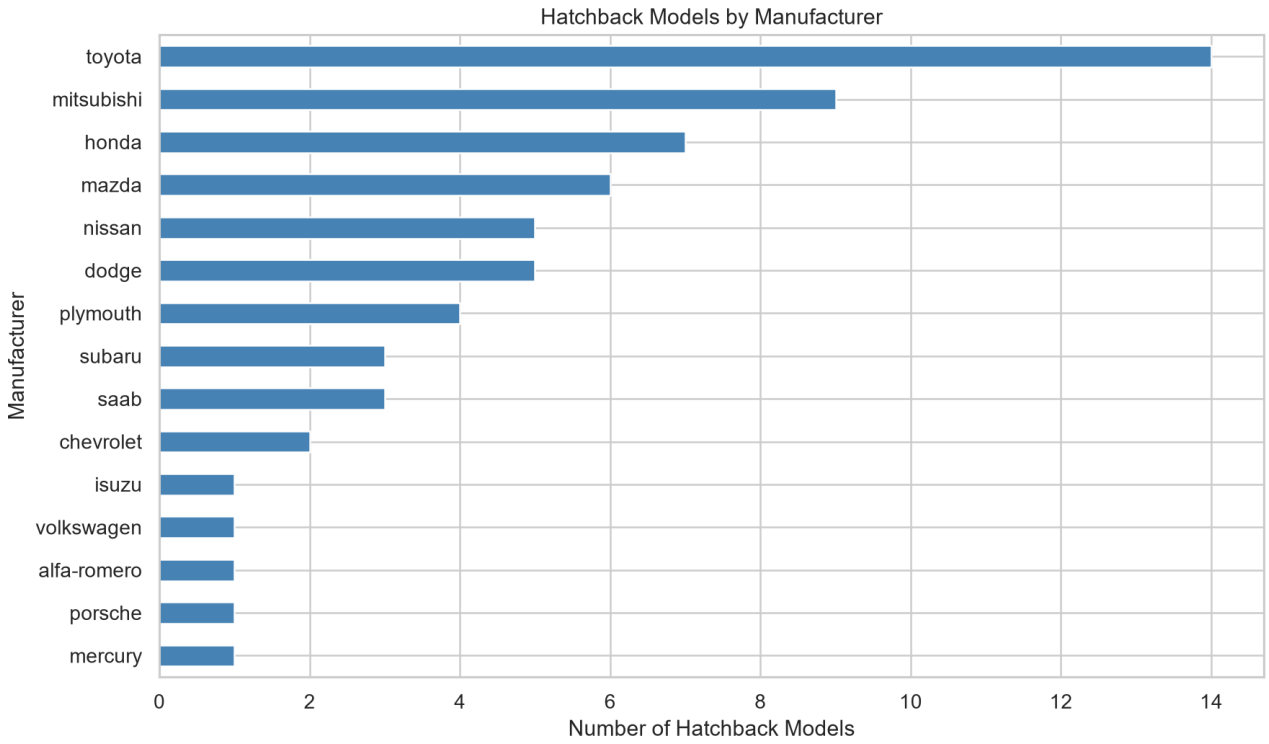
Visualisation 2: Price vs Engine Size

The scatter plot indicates a positive relationship between engine size and vehicle price. The Pearson correlation is 0.89, although premium-brand vehicles create a few high-priced outliers.



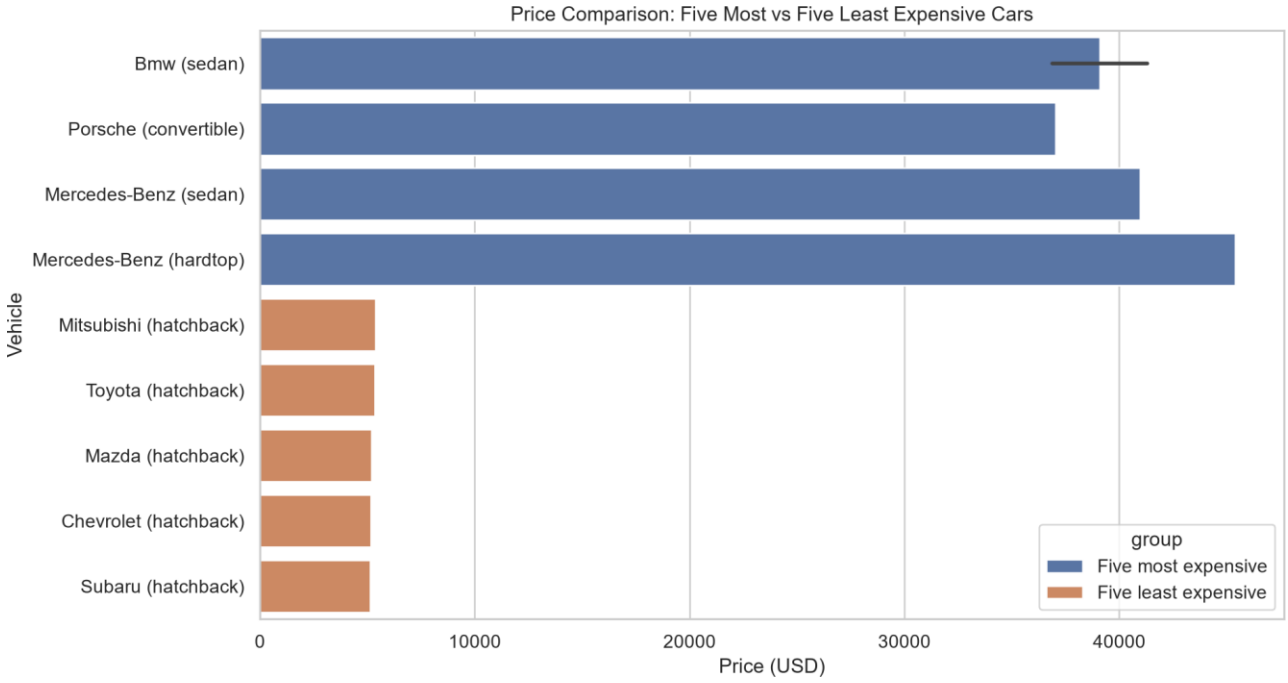
Visualisation 3: Hatchback Models by Manufacturer

There are 63 hatchbacks in the cleaned dataset. The chart shows how the hatchback category is distributed across manufacturers.



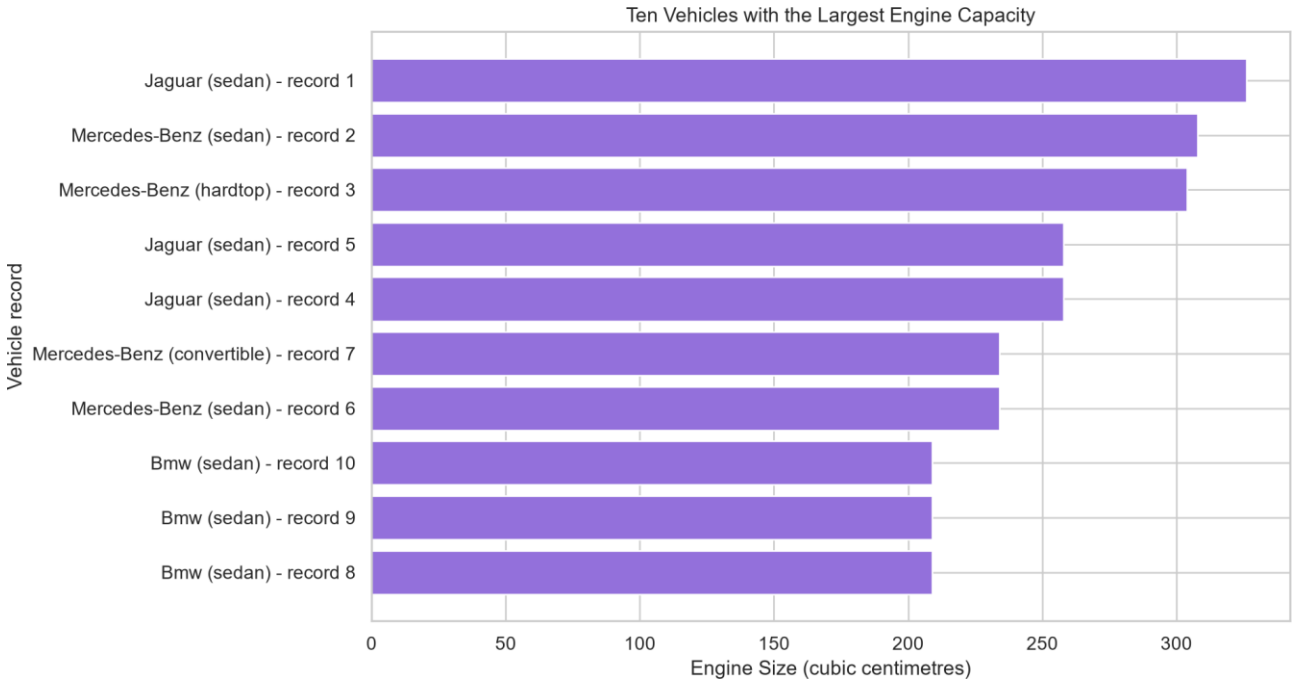
Visualisation 4: Five Most Expensive vs Five Least Expensive Cars

The most expensive vehicle costs USD 45,400, compared with USD 5,118 for the least expensive. The notebook table provides the corresponding MPG comparison.



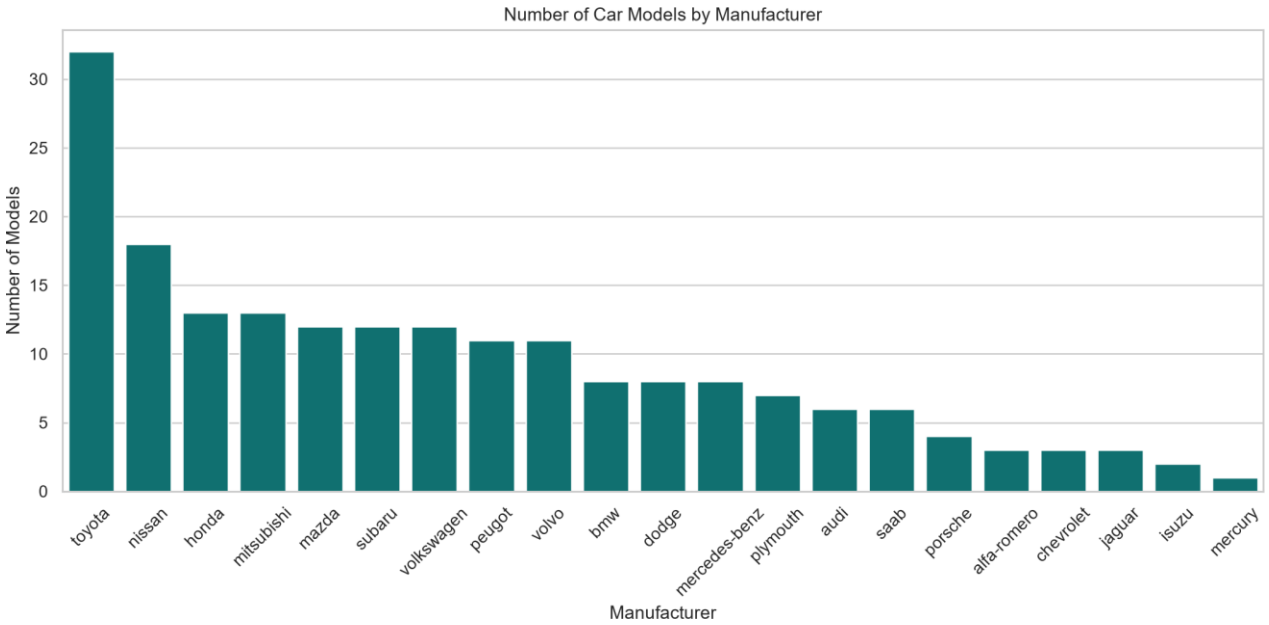
Visualisation 5: Ten Vehicles with the Largest Engine Capacity

The largest engine capacities are concentrated among a small number of manufacturers, helping explain why these vehicles tend to be in the higher-price range.



Visualisation 6: Number of Car Models by Manufacturer

Toyota has the highest number of models in the cleaned dataset. This reflects dataset representation, not real-world market share.



Key Findings

- Mercedes-Benz and BMW include the most expensive vehicles.
- Jaguar and Mercedes-Benz feature the largest engine capacities.
- Toyota has the highest number of models in the cleaned dataset.
- Chevrolet has the highest average city MPG.
- Vehicle price generally increases with engine size (correlation: 0.89).

Conclusion

The exploratory analysis identified meaningful relationships between engine size, fuel economy, manufacturer, and price. The cleaned dataset is suitable for further predictive modelling and statistical analysis. Every chart and finding in this report is generated by code in the accompanying notebook.

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